Loco 42728 (WAG-9HC) — Axle-04 Lock, 08-Aug-2026 Telemetry Evidence Report (Equus feed, 27-Jul → 10-Aug 2026)

Source: **dbo.Lotus_loco_process_signals_RDSOJson**, Vendor **Equus**, LocoId 42728. Rerun total **156,088 rows** (orig. 155,095 +993 late 10-Aug rows), 2026-07-27 10:30 → 2026-08-10 11:05. Reference: **case_42728/findings_42728.txt** (frozen) + **coupled_report_42728.html** (8 SVG charts). Physics-label rule: **loss-of-thermal-contribution state**; mechanism (cutout vs mechanical) is **not separable** from captured signals.

One-line finding: TM1_2 (axle-04 motor) slightly hot 06/08, severe overheat to **91.7°C** 07/08, flat-cold on 08/08 while the other 5 motors carried the train at 60–74°C; reported lock **08/08 19:26** falls inside the **08/08 17:25 → 09/08 14:25 telemetry gap**; **faultnum=0**, zero fault-table rows.

1. Official failure timeline (SinglePageFailureReport.xls)

When	What
08/08 19:11	arrived SBZ (NKE-CPJ, BSB/NER), WAG-9HC 42728 (BCNE), shed SYHE
08/08 19:26	starter given; LP reports AXLE LOCKED — 2nd bogie, 04 axle; train unable to move
08/08 19:32	starter put back
09/08 00:35 / 04:20 / 04:45	technician arrived / loco detached / handed over
Type / last maint.	ICMS/EPOTH; last minor SCH IA 16/05/2026; report generated 10-08-2026

2. Database reality + critical gap

Check	Result
Fault tables	Lotus_LocoFaultData / RDSOJson / Locofault → 0 rows for 42728 (fleet has 712,435 rows 07–10 Aug)
faultnum in telemetry	0 in ALL extracted rows — no fault byte captured around the lock
Old feed	Oct-2025 LotusWireless = commissioning baseline only, NOT evidence for 08/08/2026
Gap (critical)	08/08 ends 17:25:10 → resumes 09/08 14:25:11. 19:26 lock + 00:35–04:45 detach ALL inside gap
08/08 coverage	15:49:10–17:25:10 only (1416 rows); max GPS 40.2 km/h slow freight; 09/08 max 8.5, 10/08 max 8.2

3. Thermal progression — TM1_2 = axle-04 motor (axles 1–3 bogie 1, 4–6 bogie 2)

Stage	TM1_2 behaviour
06/08 14:00–15:00	+9 to +10°C above other 5 (mean delta +10.1/+9.3). First abnormal loading, 2 days prior. Also +16.7/+14.3/+9.5 episodes 14:30–15:00
07/08 10:13–10:34	OVERHEAT: 83.5 → crest 91.4–91.7 (10:20–10:27, dT +19.7) → 78.5 by 10:34 → ~71 by 10:38. Only 6-motor anomaly
07/08 19:36–19:46	TRANSITION: first loaded run after overheat — other 5 warm 42→55°C, TM1_2 flat 38.7–38.9°C. Cold state (dT<–10) sustained from ~19:46
08/08 15:49–17:25	FLAT-COLD 36.6–40.2°C while other 5 run 60–74°C (median dT –27.7, min –33.7). Axle-04 already unloaded pre-lock
09/08 14:25+	Crawl ≤8.5 km/h; TM1_2 39–42°C vs others 49–58°C — dragging/binding move post-detach

Hour	dT	Hour	dT	Hour	dT
06/13 +4.7	06/14 +10.1	06/15 +9.3	07/07 +0.5	07/08 +0.1	07/09 +1.2
07/10 +9.5 (event hr)	07/20 –13.7	08/16 –21.9	08/17 –29.6 (pre-fail)	09/15 –9.5	09/18 –18.1

4. Matched-condition tests (do NOT claim bearing cause from temperature alone)

Test	Method → result
C — axle speed (top priority)	xvist_a1_2 = axle-04. 08/08 pre-gap tracks axles 1/2/3/5 EXACTLY (still rotating, not seized). 09/08 = 3276 dead sentinel all 2392 rows (signal died post-incident). Axle-06 xvist_a3_2 = 3276 all days = artifact
B — current residual vs own f(v,a) baseline (27Jul–03Aug, lte=1)	Median I_excess: 06-Aug high bins UP (+41.1 @v40–50); 07-Aug UP across bins (+20.4/+8.3/+21.2); 08-Aug NOT elevated (–13.6/–21.7/–17.6) → load redistributed, not harder running. Small-n bins caveat
E — load redistribution (speed>20, lte=1)	07/08: TM1_2 RISES with Ip (60–80→68.2°C dT+6.5; 80–120→84.9°C dT+15.9, still coupled). 08/08: FLAT 38–40°C at EVERY Ip bucket, others 57–72°C (dT –19…–32, decoupled). Baseline dT +0.8…+2.5
D — current→temp lag 07/08 09:30–11:30	corr(dT04,Ip): τ=0 +0.667 → τ=+60s +0.853, above dT autocorr 0.849 → current leads temp (directional only; both series smooth)

Residual-distribution rerun (03-Sep-2026, stdlib only, baseline f(v,a) medians 27Jul–03Aug lte=1): $r_t = I_{obs} - \hat{I}(v,a)$, traction-active only.

Period	n	med	p10–p90	IQR	P(|r|>30)	KS_D	dT
BASE 27J–03A	26340	+0.0	–23.4…+39.3	21.3	0.205	—	+0.7
06-Aug	1751	+1.2	–19.2…+30.3	18.4	0.140	0.062	+0.9
07-Aug	893	+8.1	–5.4…+26.6	22.4	0.075	0.277	+0.5
07/08 10–11 (event)	333	+22.0	+4.1…+27.6	5.9	0.075	0.563	+15.5
07/08 14–20 (post)	374	–0.5	–11.2…+14.4	12.2	0.021	0.176	–4.7
08/08 15–17 (pre-fail)	968	–4.9	–21.5…+9.8	20.5	0.054	0.203	–21.8

Reading: shift is brief/localized to 07/08 10–11h with matching thermal +15.5°C, resets post-event, cold pre-fail –21.8°C. Supports coupled-change claims, not mechanism/generality.

5. Coupled signals 07–08 Aug (traction-active medians; bg2tm* ≈1713 V = shared DC-link, NOT per-axle)

Slice	n	Ip med (p90)	TM1_2	other5	dT
07/08 30–40 km/h	69	65.7 (74.5)	67.5	61.3	+6.2 (hot+coupled)
BASE 30–40	2415	58.8 (127.7)	57.0	56.1	+0.9
08/08 40–50	13	120.6 (126.7)	39.9	69.3	–29.5 (cold, 5 carry; small-n)
BASE 40–50	2435	60.1 (154.8)	59.7	57.5	+2.2
07/08 10–11 event hr	—	61.0 (70.3)	66.9	61.0	+5.9
08/08 15–17 pre-fail	—	23.2 (58.3)	39.1	64.8	–25.7

07/08 10:01–10:34 aligned: 12–57 km/h, Ip 42–92 A, TM1_2 63→86→91.7°C; peak 10:27 with speed sag to ~40 km/h, Ip ~10 A; 10:34 stopped (v=0, DC-link ~1520 V). 08/08 15:49–17:25: TM1_2 flat 36.6→40.2 whole session; others 38→74; accel to 60–73 km/h, Ip 130–144 A peak. Energy counter rate climbs directionally (05/08 1.0× → 06/08 2.1× → 07/08 1.55× → 08/08 4.3× on 87 running-min; cumulative counter, units unverified — no 12665 kWh/min quote).

6. Task-7 reconstruction 07/08 19:15→19:50 (462 rows, 1-s; faultnum=0, all cutout bits 0)

Segment	TM1_2	other5	dT	Ip	v
Before 19:36 (idle)	39.1	38.8	+0.3	9.8	0
19:36–19:46 (loaded)	38.9	44.9	–6.1	20.4	27.6
After 19:46	38.7	55.1	–16.5	9.6	29.3

19:15–19:35 idle (all six drift 39.4→39.0; 15-s shunt 19:27:42–55, all axles together). 19:36:42 first loaded run (xang→50, Ip→81 A): others 42.8→47.1, TM1_2 flat. 19:40:45→19:46:09 ~5.4-min gap (others 47→55 inside gap). 19:46:32–52 brake demand. Axle-04 rotates at train speed throughout. H-A sensor fault NOT supported (tracks ambient at idle, read 91.7 that morning, smooth ~38.8 under load — live unloaded channel, unlike 3276/76.0 stuck artifacts). H-B cutout consistent but no bit evidence (feed exposes no per-motor bit). H-C friction fits morning overheat + residual. "19:46 dT<–10" is others reaching ~55°C, not a second axle-04 event — decoupling present from 19:36.

7. What we can / cannot claim + artifacts

#	Claim	Verdict
1	Thermal precursor (06 +9–10, 07 91.7 dT+19.7, 08 dT–27.7 cold)	YES
2	Electrical/thermal coupling change (E: follows 07/08, flat 08/08)	YES
3	Aggregate electrical abnormality (B: elevated 06+07 at matched v,a)	YES
4	Early-warning potential (~1 day; transition 07/08 19:36–19:46)	YES
5	Causal mechanism (cutout vs mechanical)	NO — bits never changed; failure in gap
6	Bearing-specific signature (BPFO/BPFI)	NO — no kHz waveform
7	General predictive capability	NO — n=1; needs other cases + healthy controls

Exclusions: xvist_a3_2 = 3276 all days; xtempmotor3_2 = 76.0°C clip ~100% (both axle-06 artifacts, excluded from means); bg1tm*/bg2tm*/bur* only from 05-Aug; Oct-2025 = baseline only. Negative results retained: bbur/bstb = 0 on 08/08 run; aggregate current not monotonic; per-axle current not recorded — inference rests on per-axle temp + aggregate current/speed/energy.

8. Conclusion & next step

42728 shows a measurable pre-lock thermal + aggregate-electrical behavioural change with TM1_2 decoupling ~1 day before the reported lock — an **electrical/thermal precursor hypothesis**, not a bearing diagnosis. Chain (Shang as principle, not algorithm): expected electrical behaviour → residual → thermal response → cross-motor deviation. Next: run this exact TM-abnormality → excursion → load-response pattern on the other 2025 axle-lock cases and healthy controls. Case analysis DONE; predictive conclusion OPEN (closeout 03-Sep-2026).
Sources: SinglePageFailureReport.xls; telemetry_42728_2026_rds.json.csv (Equus); scripts verify_axle4/timeline_check/analyze_2026_*/test_B_plus/test_E/test_F_transition/coupled_analysis/make_coupled_report/task7_window/task7_report/extract_42728_full+resume/residual_distribution_42728.py; charts: coupled_report_42728.html.